

$$\text{Arc Distance} = 2\pi rX$$

The diagram illustrates a 3-DOF parallel robot mechanism. It features a central vertical axis and a horizontal base. The mechanism consists of three legs, each with a base joint (CL, LC, CR) and a top joint (FL, LF, FR). The legs are connected to a moving platform (LF, LR, RR). The diagram includes several geometric parameters and labels:

- Instantaneous Center of Rotation (ICR):** Located at the origin of the coordinate system.
- Angles:** θ_L , θ , and θ_R are indicated at the joints.
- Distances:** L and $2L$ are indicated on the right side.
- Formulas:**
 - $[2\pi \text{ rLC}] \text{ dLC}$
 - $[rLC-T] \text{ rCL}$
 - $[\sqrt{(rCL^2 + L^2)}] \text{ rSL}$
 - $[L/\tan\theta] \text{ rLC}$
 - rLS
 - $[rLC+T] \text{ rCR}$
 - rSR
- Dimensions:** T and $2T$ are indicated at the bottom.

At the bottom, the following formulas are provided:

$$\theta_L = \text{atan}(L/(L/\tan\theta)-T)$$

$$\theta_R = \text{atan}(L/(L/\tan\theta)+T)$$